Dear Dr. Holmboe's Team,

I am writing to express my interest in the position of Research Support Assistant, focusing on the study of executive and attentional functions. My academic and research experiences have deeply rooted my interests in neuroscience, providing me with a broad knowledge of brain biology and the physics principles underlying advanced imaging and physiological recording technologies. With experience in multi-disciplinary lab environments, I have developed strong project management and research skills, along with a high awareness of responsibilities in laboratory setting. Additionally, I possess practical skills in data analysis, including mathematical and computational tools specialised for processing neural data, which I believe are highly transferable to this position.

During my Master's study of Biomedical Engineering, specialising in Neurotechnology, I acquired a board knowledge of neuroscience, understanding brain mechanisms from the molecular level to higher-level functions such as attention. My experience with neural data programming, particularly decoding arm movement trajectories from single neuron and multiple unit recordings, has enhanced my skills in neural data processing. I am also well-versed in data protection and maintain tidy data-keeping habits.

My diverse laboratory experience, including electronics, neurophysiological recording, and advanced medical equipment operation, has refined my research skills. Hardware-wise, I prototyped a wearable device that records and translates EMG signals into audio signals for biofeedback of muscle control. Additionally, I have experience operating ultrasound and advanced microscopy systems. By designing experiments to calibrate the GUI software with the hardware power generated through the ultrasound transducer and conducting pre-clinical testing to write an end-user guide, I gained valuable hands-on experience that would be applicable to this position. My previous role s a research assistant improved my decision-making and effectively communication with the team, supervisors and other collaborated partners. The independence in designing and developing projects improved my problem-solving and organisational skill. My experience with cell culture has equipped me with the ability to follow and develop experimental protocols, which require precise operation, troubleshooting, and frequent reviewing to ensure project success. I believe that my knowledge and relevant lab experience would make me a quicker learner for the techniques applied in this position.

I have developed my interpersonal skills through various working and volunteering experiences, which are transferable to clinical work with participants. Specifically, I volunteered at a primary school with children with learning difficulties, where I continuously refined my approaches to deliver information and capture their attention during one-to-one sessions. Through active listening and patiently following with their pace, I improved my ability to interact with young children. I am eager to enhance my skills in participant-centered research and I am highly motivated to learn for the sake of my future research journey.

My interest in developmental psychology and developmental cognitive neuroscience was sparked by multiple influences. In this era of information overload, where high-demanding brain work dominates the job market, the influx of fragmented information significantly impacts mental health, particularly in relation to attentional functions. Personally, I am interested in and actively seeking solutions in daily life to 'train' my brain, counteracting the negative influences while benefiting from modern technologies. I believe that just like how physically training our muscle, there are practical approaches to enhance our mental strength. Understanding the mechanisms of the cognitive processes would be a significant aspect of developing treatments for various diseases. I am keen to contribute to investigating the fundamental principles of cognitive processes, including discovering genetic, environmental and developmental factors. Additionally, I feel a responsibility to raise public awareness about protecting and maintaining mental energy for optimal use.

I believe my interests and abilities aligns well with the requirements of this position. The opportunity to join your active team would be invaluable for enhancing both my practical and soft skills in research, as well as preparing me for pursuing a Ph.D. I am excited about the prospect of contributing to and learning from your research team.

I look forward to hearing from you and discussing my research interests and background in more detail.

Yours sincerely,

Chuqi